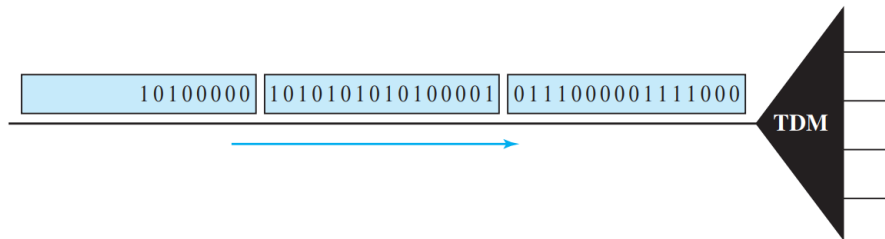


CSE 320 Quiz 3

Marks - 15 Time - 25 min

ID	
Name	

1. Five channels, each with a **50 kHz** bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of **100 Hz** between the channels to prevent interference? **[3]**
2. Explain the difference between synchronous and statistical TDM **[2]**
3. The figure below shows a demultiplexer in a synchronous TDM. If the input slot is **16 bits** long (no framing bits), what is the bit stream in each output if one bit is taken as a unit? Also what will be the output if two bits are taken as a unit? The bits arrive at the demultiplexer as shown by the arrows. **[7]**



4. We have a digital medium with a data rate of **11 Mbps**. How many **64 kbps** voice channels can be carried by this medium if we use **DSSS** with the Barker sequence? **[3]**